Problem 4.5

(Difficulty: 2)

4.5 A 0.3 m by 0.5 m rectangular air duct carries a flow of 0.45 $\frac{m^3}{s}$ at a density of 2 $\frac{kg}{m^3}$. Calculate the mean velocity in the duct. If the duct tapers to 0.15 m by 0.5 m size, determine the mean velocity in this section if the density is 1.5 $\frac{kg}{m^3}$ in this section.

Given: Duct size: $w_1 = 0.3\ m$; $L_1 = 0.5\ m$; $w_2 = 0.3\ m$; $L_2 = 0.5\ m$. Density: $\rho_1 = 2\ \frac{kg}{m^3}$; $\rho_2 = 1.5\ \frac{kg}{m^3}$.

Volumetric flow rate: $Q = 0.45\ \frac{m^3}{s}$.

Find: The mean velocity V_1 and V_2.

Assumption: The density is constant

Solution: Use the continuity equation

$$\dot{m} = \rho \quad V \quad A$$

For the duct entrance, we have for the flow area:

$$A_1 = w_1 L_1 = 0.3\ m \times 0.5\ m = 0.15\ m^2$$

The mean velocity can be calculated:

$$V_1 = \frac{Q}{A_1} = \frac{0.45\ \frac{m^3}{s}}{0.15\ m^2} = 3\ \frac{m}{s}$$

For the tapered section, the mass flow rate is the same as:

$$\dot{m} = \rho_1 V_1 A_1 = 2\ \frac{kg}{m^3} \times 3\ \frac{m}{s} \times 0.15\ m^2 = 0.9\ \frac{kg}{s}$$

The flow area is:

$$A_2 = w_2 L_2 = 0.15\ m \times 0.5\ m = 0.075\ m^2$$

So the mean velocity in the tapered section is:

$$V_2 = \frac{\dot{m}}{\rho_2 A_2} = \frac{0.9\ \frac{kg}{s}}{1.5\ \frac{kg}{m^3} \times 0.075\ m^2} = 8\ \frac{m}{s}$$

Problem 4.9

(Difficulty: 2)

4.9 A pipeline 0.3 m in diameter divides at a Y into two branches 200 mm and 150 mm in diameter. If the flow rate in the main line is 0.3 $\frac{m^3}{s}$ and the mean velocity in the 200 mm pipe is 2.5 $\frac{m}{s}$, determine flow rate in the 150 mm pipe

Given: The diameter for the main line: $D_1 = 0.3\ m$. The diameter for the two branches: $D_2 = 200\ mm$ and $D_3 = 150\ mm$. The flow rate in the main line is: $q_1 = 0.3\ \frac{m^3}{s}$. The mean velocity in the 200 mm pipe: $V_2 = 2.5\ \frac{m}{s}$

Find: The flow rate q_3 in the 150 mm pipe.

Solution:

From the continuity equation we have:

$$0 = \frac{\partial}{\partial t}\int_{CV} \rho d\forall + \int_{CS} \rho \bar{V} \cdot d\bar{A}$$

For steady flow there is no change with time and we have:

$$0 = \int_{CS} \rho \bar{V} \cdot d\bar{A}$$

or

$$\rho V_1 A_1 = \rho V_2 A_2 + \rho V_3 A_3$$

Since the density of water is constant

$$V_1 A_1 = V_2 A_2 + V_3 A_3$$

$$Q_1 = Q_2 + Q_3$$

where

$$Q_1 = V_1 A_1$$

$$Q_2 = V_2 A_2$$

$$Q_3 = V_3 A_3$$

Thus

$$Q_2 = V_2 A_2 = \frac{\pi}{4} D_2^2 V_2 = 2.5\ \frac{m}{s} \times \frac{\pi}{4} \times (0.2\ m)^2 = 0.0785\ \frac{m^3}{s}$$

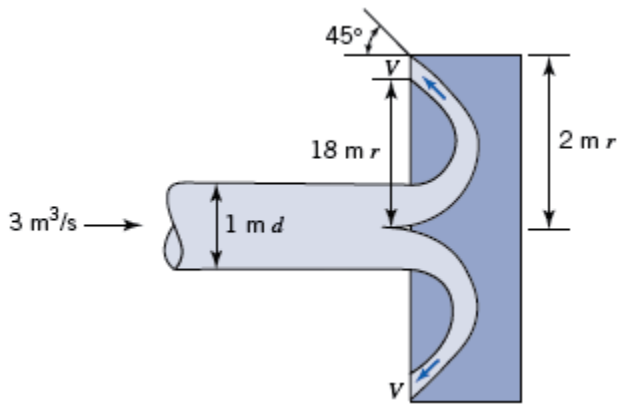
So we can find the flow rate in the 150 mm pipe:

$$Q_3 = Q_1 - Q_2 = 0.3\ \frac{m^3}{s} - 0.0785\ \frac{m^3}{s} = 0.222\ \frac{m^3}{s}$$

Problem 4.10

(Difficulty: 2)

4.10 Find V for this mushroom cap on a pipeline.



Given: All the other parameters are shown in the figure.

Find: The velocity V.

Assume: The density is constant

Solution: From the continuity equation we have:

$$0 = \frac{\partial}{\partial t}\int_{CV} \rho d\forall + \int_{CS} \rho \bar{V} \cdot d\bar{A}$$

For steady flow there is no change with time and we have:

$$0 = \int_{CS} \rho \bar{V} \cdot d\bar{A}$$

The density is constant we have:

$$Q_{in} = Q_{out}$$

Also we can get:

$$Q_{out} = V \cos 45° \, A_{out}$$

The outlet area is:

$$A_{out} = \pi(r_2^2 - r_1^2) = \pi[(2\,m)^2 - (1.8\,m)^2] = 2.39\,m^2$$

The velocity is calculated to be:

$$V = \frac{Q_{out}}{A_{out} \cos 45°} = \frac{3\ \frac{m^3}{s}}{2.39\ m^2 \times \cos 45°} = 1.78\ \frac{m}{s}$$

Problem 4.11

[Difficulty: 1]

4.11 You are trying to pump stormwater out of your basement during a storm. The basement is 20 ft× 30 ft and the pump extracts 27.5 gpm. The water level in the basement is dropping 4 in./hr. Determine the flow rate (gpm) from the storm into the basement.

Given: Data on filling of a basement during a storm

Find: Flow rate of storm into basement

Solution:

This is an unsteady problem if we choose the CS as the entire basement

Basic equation $$\frac{\partial}{\partial t}M_{CV} + \sum_{CS}\left(\rho\cdot\vec{V}\cdot\vec{A}\right) = 0$$

Assumptions: 1) Incompressible flow

Given data: $Q_{pump} = 27.5\cdot gpm$ $\quad \frac{dh}{dt} = 4\cdot\frac{in}{hr}$ $\quad A = 30\cdot ft\cdot 20\cdot ft$

Hence $$\frac{\partial}{\partial t}M_{CV} = \rho\cdot A\cdot\frac{dh}{dt} = \sum_{CS}\left(\rho\cdot\vec{V}\cdot\vec{A}\right) = \rho\cdot Q_{storm} - \rho\cdot Q_{pump}$$

where A is the basement area and dh/dt is the rate at which the height of water in the basement changes.

or $$Q_{storm} = Q_{pump} - A\cdot\frac{dh}{dt}$$

$$Q_{storm} = 27.5\cdot\frac{gal}{min} - 30\cdot ft\times 20\cdot ft\times\left(\frac{4}{12}\cdot\frac{ft}{hr}\right)\times\frac{7.48\cdot gal}{ft^3}\times\frac{1\cdot hr}{60\cdot min}$$

Data on gals from Table G.2

$$Q_{storm} = 2.57\cdot gpm$$

Problem 4.12

(Difficulty: 3)

4.12 A cylindrical tank, of diameter $D = 150\ mm$, drains through an opening, $d = 5mm$, in the bottom of the tank. The speed of the liquid leaving the tank is approximately $V = \sqrt{2gy}$ where y is the height from the tank bottom to the free surface. If the tank is initially filled with water to $y_0 = 0.4m$, determine the water depths at $t = 60\ sec$, $t = 120\ sec$, and $t = 180\ sec$. Plot $y(m)$ versus t for the first 180 sec.

Given: Data on draining of a tank

Find: Depth at various times. Plot of depth versus time.

Assumption: (1) Uniform flow

(2) Incompressible flow

(3) Neglect air density

Solution:

Basic equation:

$$0 = \frac{\partial}{\partial t}\int_{CV} \rho d\forall + \int_{CS} \rho \vec{V} \cdot d\vec{A}$$

Treaing the tank as the CV the basic equation becomes:

$$\frac{\partial}{\partial t}\int_0^y \rho \cdot A_{tank} dy + \rho \cdot V \cdot A_{opening} = 0$$

$$\rho \cdot \frac{\pi}{4} \cdot D^2 \cdot \frac{dy}{dt} + \rho \cdot \frac{\pi}{4} \cdot d^2 \cdot V = 0$$

Using

$$V = \sqrt{2gy}$$

and simplifying the equation:

$$\frac{dy}{dt} = -\left(\frac{d}{D}\right)^2 \sqrt{2g} y^{\frac{1}{2}}$$

Seperating variables

$$\frac{dy}{y^{\frac{1}{2}}} == -\left(\frac{d}{D}\right)^2 \sqrt{2g} dt$$

Integrating the equation:

$$2\left(y^{\frac{1}{2}} - y_0^{\frac{1}{2}}\right) = -\left(\frac{d}{D}\right)^2 \sqrt{2g}t$$

$$y^{\frac{1}{2}} = -\left(\frac{d}{D}\right)^2 \sqrt{\frac{g}{2}}t + y_0^{\frac{1}{2}}$$

$$y^{\frac{1}{2}} = y_0^{\frac{1}{2}}\left(1 - \left(\frac{d}{D}\right)^2 \sqrt{\frac{g}{2y_0}}t\right)$$

So we have:

$$y = y_0\left(1 - \left(\frac{d}{D}\right)^2 \sqrt{\frac{g}{2y_0}}t\right)^2$$

As we know:

$$y_0 = 0.4\ m$$

$$\frac{d}{D} = \frac{5\ mm}{150\ mm} = \frac{1}{30}$$

At $t = 60\ s$, we have:

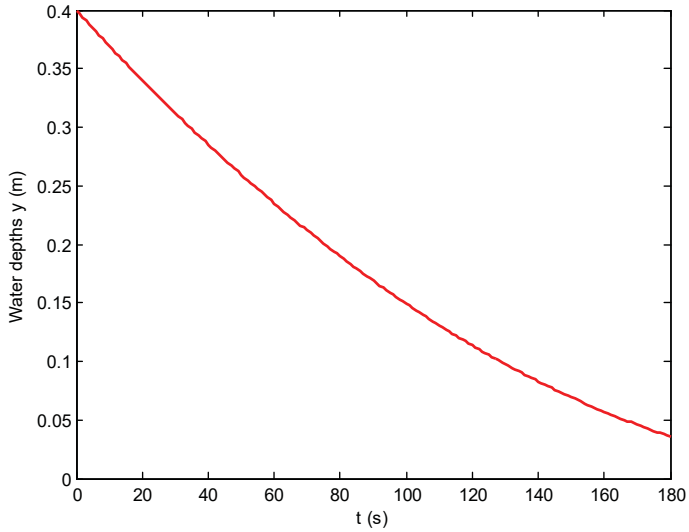
$$y = 0.235\ m$$

At $t = 120\ s$, we have:

$$y = 0.1137\ m$$

At $t = 180\ s$, we have:

$$y = \ 0.0359\ m$$


0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
Water depths y (m)
0
20
40
60
80
100
120
140
160
180
t (s)